

AthleteQuest AI

Your Personal AI Athlete Twin

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1. Project Description & Name

Project Name

AthleteQuest AI: Your Personal AI Athlete Twin

Project Description

AthleteQuest AI is an AI-powered coaching app that helps young athletes improve their performance, build healthy habits, and stay motivated.

The app creates an AI Athlete Twin, a personalized digital model that learns from an athlete's goals, training history, performance records, recovery habits, and motivation style.

Unlike traditional fitness apps, AthleteQuest continuously adapts coaching recommendations and predicts future performance based on each athlete's progress.

Everything runs **100% on the device** — there are no accounts, no servers, and no cloud. A child's data never leaves their phone. The goal is to make personalized coaching accessible, fun, private, and engaging for all young athletes.

2. The Problem

Many young athletes struggle to improve because they do not have access to personalized coaching. Common challenges include:

- Expensive coaching programs
- Generic training plans
- Losing motivation after a few weeks
- Difficulty tracking progress
- Lack of guidance on recovery and nutrition

Every athlete develops differently, but most training apps treat all athletes the same. As a result, athletes may become frustrated, stop training, or fail to reach their goals.

3. Our Approach

AthleteQuest AI creates an AI Athlete Twin for every athlete. The Athlete Twin stores:

- Age
- Height
- Weight
- Sport
- Personal records
- Training history
- Recovery information
- Motivation preferences

The app analyzes this information — all on the device — and creates:

- Personalized training plans
- Dynamic daily quests
- Performance predictions
- Coaching recommendations

The Athlete Twin becomes smarter as more information is collected.

4. Impact

AthleteQuest AI has the potential to create positive impacts in several areas.

Health

Encourages regular exercise and physical activity.

Motivation

Uses rewards, streaks, badges, and character progression to keep athletes engaged.

Confidence

Helps athletes see measurable improvement over time.

Accessibility

Provides coaching support to athletes who may not have access to personal trainers — for free, on any phone.

Education

Teaches healthy training habits, recovery strategies, and goal setting.

Potential users include:

- School athletes
 - Recreational athletes
 - Community sports teams
 - Physical education programs
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5. AI Integration

Artificial Intelligence is the core technology behind AthleteQuest AI — and it all runs on the device.

AI Coach

The AI Coach lets athletes ask questions in plain language — by typing or by voice. Examples:

- How can I run faster?
- What should I eat before a race?
- Why am I feeling tired?

Every answer is computed from the athlete's own Athlete Twin, so advice is personalized — not generic.

Athlete Twin

The Athlete Twin stores and analyzes athlete data. The app learns:

- Strengths
- Weaknesses
- Training habits
- Recovery habits
- Motivation patterns

Performance Prediction

The app estimates future outcomes. For example, "Can I run a 6-minute mile by September?" It predicts:

- Current trajectory
- Probability of success
- Recommended improvements

Motivation Engine

The app learns what motivates each athlete and adjusts recommendations accordingly.

6. Key Features

Personalized Athlete Twin

Digital model of the athlete.

AI Coach

Interactive coaching assistant — type or speak to it.

Performance Forecasting

Predicts future athletic performance.

Dynamic Training Plans

Adjusts plans automatically based on progress.

Daily Quests

Generates personalized challenges.

RPG Progression

Athletes gain XP and level up.

Character Customization

Unlock outfits, accessories, and rewards.

Parent Dashboard

Allows parents to monitor progress.

Streak Protection

Encourages consistency without punishing missed days.

7. How the AI Works

This project does **not** train a large language model and does **not** call any cloud AI service. The intelligence runs entirely on the device, using three on-device parts:

Rule-Based Coaching Engine

A deterministic engine routes each question and builds its answer from the Athlete Twin (sport, goals, trends, recovery). Because it is rule-based, responses are predictable and safe for kids — there is no risk of a model inventing unsafe advice.

Forecasting Model

A linear-regression model is trained on the athlete's own performance logs and re-trains every time a new result is added, producing the trend and goal-probability forecasts.

On-Device Voice

Voice input uses an on-device speech-to-text model (Whisper tiny, via transformers.js, a one-time ~40 MB download). Audio is transcribed locally and never leaves the device.

Built-in Coaching Rules

The coaching engine is built to:

- Act as a youth sports coach
- Provide age-appropriate guidance
- Encourage healthy habits
- Promote recovery and safety
- Stop training advice and refer to a parent/physician when injury is reported
- Avoid diet, supplement, and other unsafe recommendations

8. Dataset

AthleteQuest uses the athlete's own data, stored locally on the device. Sources include:

User Profile Data

Collected from the onboarding survey — e.g. age, height, weight, sport.

Performance Data

E.g. sprint times, mile/swim times, personal records.

Training Data

E.g. completed workouts, training frequency, daily activity.

Recovery Data

E.g. sleep, energy levels, recovery status.

Knowledge Sources

The coaching engine is guided by public sports-science concepts, general fitness guidance, healthy training principles, and nutrition best practices. No personal medical information is required, and **all athlete data stays in on-device storage** — nothing is uploaded.

9. Technical Challenges

Challenge 1

Creating age-appropriate coaching.

Solution: Safety rules and guardrails ensure responses remain suitable for young athletes.

Challenge 2

Keeping users motivated.

Solution: Gamification including XP, levels, quests, badges, and rewards.

Challenge 3

Making recommendations personalized.

Solution: The Athlete Twin stores historical athlete information and provides context to the coaching engine.

Challenge 4

Preventing overtraining.

Solution: Recovery scoring and rest recommendations are built into the coaching system.

Challenge 5

Protecting kids' privacy.

Solution: The whole app was built to run offline — no accounts, no servers, no data collection — so there is nothing to leak.

10. Technology Stack

The entire app is a single self-contained HTML file that runs **100% offline** in any modern browser — no backend, no servers, no accounts, no cloud.

App

- Single-file HTML, CSS, and vanilla JavaScript
- Installable Progressive Web App that works fully offline
- No framework and no build step

Data Storage

- Browser localStorage — all data stays on the device
- No database and no remote storage

On-Device Intelligence

- Rule-based coaching engine over the Athlete Twin (deterministic, kid-safe)
- Linear-regression forecasting, re-trained locally on each new result
- On-device speech-to-text (Whisper tiny via transformers.js) for voice chat

Accounts, Hosting & Analytics

- No accounts or sign-in
- No analytics or tracking of any kind
- Distributed as a static web app — nothing about a child's data leaves their device

What changed from the early plan

An earlier design considered cloud services — Clerk (auth), Supabase (database), OpenAI GPT (AI), Vercel (hosting), and PostHog (analytics). These were intentionally removed so the app could be 100% offline and private by design.

11. Responsible AI & Privacy

The coach should always:

- Encourage healthy habits
- Encourage recovery
- Encourage hydration
- Encourage goal setting

The coach should never:

- Recommend dangerous training
- Encourage unhealthy dieting
- Promote overtraining

Privacy by Design

Because the app is fully on-device, there are no accounts, no cloud, and no data collection — a child's information never leaves their phone. If an injury is reported, the app stops giving training advice and refers the athlete to a parent and a physician. Safety and privacy are core design principles.

12. Future Improvements

Future versions could include:

- Optional on-device LLM chat (e.g. Google Gemma) — still fully offline, no cloud
 - Video analysis
 - Running form analysis
 - Motion tracking
 - Team challenges
 - Wearable integrations
 - Voice coaching
 - AR training experiences
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Conclusion

AthleteQuest AI combines on-device artificial intelligence, personalized coaching, health education, and gamification into a single, fully offline app.

By creating a personalized Athlete Twin that lives entirely on the athlete's own device, the system helps young athletes train smarter, stay motivated, and achieve their goals — while keeping their data private and promoting healthy habits and responsible AI use.